AETHENA INSTITUTE

"TONE" INTERPRETOR

Authors: Michael Sommer & CoPilot
Date original: 06.02.2026 / Date edit:14.05.2026

SECTION 1 – TITEL

TONE: A Semantic Mediation Architecture for Translating Human Intent into Machine-Executable Actions

---

SECTION 2 – FIELD

This invention relates to:
- **human–computer interaction**
- **semantic interpretation systems**
- **identity mediation**
- **digital governance**
- **ambiguity resolution**
- **intent-to-action translation**

More specifically, it concerns a configurable, multi-layer architecture that mediates between human intent and digital system execution through:
- a **semantic translator** (Personal Digital Twin, PDT)
- a **governance substrate** (AURORA)
- a **responsibility and accountability support layer** (s.net)

All components operate **subject to applicable law, organizational policy, and configurable risk thresholds**.

SECTION 3 – BACKGROUND

Digital systems typically require structured, explicit inputs, while humans' express intent in natural, ambiguous, and context-dependent forms. This mismatch can result in:
- misinterpretation
- misclassification
- over-collection of data
- loss of operator sovereignty
- inconsistent system behaviour
- context collapse
- identity fragmentation
- unintended actions

Existing systems often lack an integrated, auditable mechanism that combines semantic translation, governance, privacy-preserving identity mediation, and configurable accountability support. As a result, misunderstandings can lead to errors, privacy harms, and loss of user control.

SECTION 4 – SUMMARY

TONE is a configurable semantic mediation architecture that translates natural human intent into machine-executable actions through a structured, multi-layer pipeline. The architecture is designed to emphasize **transparency, provenance, human oversight, and policy-bound automation**.
The architecture comprises:

1. **Operator Layer** – Receives natural, unstructured human intent in multiple modalities.
2. **PDT Layer (Translator Layer)** – Interprets intent and generates structured meaning units ("tones"); performs identity mediation, ambiguity detection, and produces provenance and confidence metadata.
3. **AURORA Layer (Governance Substrate)** – Validates tones for clarity, scope, safety, and semantic consistency using versioned rules and policy checks; coordinates clarification and escalation workflows.
4. **System Layer (Execution Layer)** – Executes only validated and authorized tones within the scope and constraints defined by the governance substrate and applicable policies.

5. **Responsibility and Accountability Layer (s.net)** – Records provenance, decision rationale, and audit logs; supports allocation of accountability according to law and organizational policy and orchestrates escalation to human actors when required.

Together, these layers aim to improve interpretation accuracy, reduce unintended actions, and provide auditable decision traces while preserving operator rights and privacy.

SECTION 5 – DETAILED DESCRIPTION

5.1 Overview

TONE is a multi-layer semantic mediation architecture that converts human intent into machine-executable actions through a structured pipeline. Each layer performs distinct functions and is **configurable**; novelty arises from their interaction, sequencing, and governance under policy and legal constraints.

Layers include:

- Operator Layer
- PDT Layer (Personal Digital Twin)
- BDT Layer; Link-Clusters; National/Continental/Global Systems; Predictability Council; Space Systems (where applicable and subject to governance)
- AURORA Layer (Governance Substrate)
- System Layer (Execution)
- Responsibility & Accountability Layer (s.net)
- Responsible, Moral & Ethical Layer; Compliance & Legal Layer

Implementation requires documented governance processes for rule updates, audits, and legal review.

5.2 Operator Layer

Function: The Operator Layer is the human source of intent. Intent may be expressed through natural language, gesture, tone of voice, contextual cues, implicit meaning, incomplete statements, or ambiguous references.

Design principles and behaviour:

- The system **accepts unstructured input** and provides prompts, confirmations, and clarification workflows to help operators supply sufficient, lawful, and accurate information.
- Operators retain responsibility for the lawfulness and intent of their requests; the system assists by surfacing potential issues and requesting clarification where needed.
- Provenance metadata is captured at input to support later review and dispute resolution.

Limitations: The Operator Layer does not absolve operators of legal obligations; it supports clearer expression and auditability.

5.3 PDT Layer (Personal Digital Twin)

Overview: The PDT Layer mediates between unstructured human input and system requirements by producing a structured, governance-ready representation called a **tone**. The PDT is configurable and operates **subject to applicable law and organizational policy**.

5.3.1 Intent Capture

- **Function:** Accepts raw operator input in multiple modalities (natural language, gesture, voice, contextual signals).
- **Clarification workflow:** Prompts for confirmation or additional detail when inputs are incomplete, ambiguous, or potentially unlawful.
- **Design principle:** The system **assists** operators in expressing intent clearly; it does not remove operator responsibility for unlawful or harmful requests.
- **Logging:** Captures provenance metadata for each capture event to support later review.

5.3.2 Semantic Interpretation

- **Function:** Transforms captured input into a structured meaning unit (**tone**) that includes **operator intent; contextual metadata; scope; constraints; identity scope; required permissions; provenance**.
- **Configurability:** Interpretation rules are versioned and configurable; outputs include confidence scores and provenance to support human review.

• **Bias mitigation:** Interpretation components include bias-monitoring hooks and explainability metadata to surface potential disparate impacts.

5.3.3 Identity Mediation
• **Function:** Produces **scoped identity assertions** sufficient for authorization and auditing while minimizing personal data disclosure.
• **Privacy controls:** Uses data minimization, purpose limitation, and privacy-preserving techniques (for example, selective disclosure or cryptographic proofs where appropriate).
• **Operator transparency:** Operators are informed which identity attributes are requested and may review or revoke consent where required by policy.

5.3.4 Ambiguity Detection
• **Function:** Detects low-confidence or conflicting interpretations and flags tones for clarification or escalation.
• **Behaviour:** When ambiguity is detected, the PDT issues structured clarification prompts, suggests constrained rewrites, or escalates to human review according to configured policies and risk thresholds.
• **Human-in-the-loop:** Final disambiguation may require operator confirmation or human adjudication depending on risk level.

5.3.5 Output
• **Function:** Emits a governance-ready tone that includes interpretation metadata, confidence scores, provenance, and any required authorization tokens.
• **Constraints:** Only tones that meet configured validation criteria are forwarded to the governance substrate; others follow clarification or escalation workflows.
Safeguards: The PDT emphasizes transparency, provenance, and configurable human oversight rather than absolute automation. Implementation must include privacy, fairness, and legal review processes.

5.4 AURORA Layer (Governance Substrate)
Overview: AURORA is a configurable governance substrate that validates interpreted tones against versioned interpretive rules, policy checks, and risk thresholds, and coordinates human-in-the-loop escalation where required by policy or law.

5.4.1 Rule Enforcement
• **Function:** Applies versioned interpretive rules and policy checks to determine whether a tone meets clarity, scope, safety, and authorization requirements.
• **Traceability:** Records rule version, rule author, and rationale for each validation decision.

5.4.2 Drift Mitigation
• **Function:** Monitors semantic consistency across sessions, devices, contexts, and time.
• **Behaviour:** Flags potential drift for review and, where configured, initiates rollback, revalidation, or human adjudication rather than asserting absolute prevention.

5.4.3 Ambiguity Resolution
• **Function:** For intentionally open-ended or low-confidence tones, AURORA routes tones to a configurable clarification workflow that may block execution, request clarification, suggest constrained rewrites, or escalate to human review based on risk and policy thresholds.
• **Design principle:** Prioritizes human oversight for high-risk or legally sensitive interpretations.

5.4.4 Safety and Operator Sovereignty
• **Function:** Ensures that validated tones do not expand operator intent beyond declared scope and that operator preferences and constraints are preserved.
• **Limitations:** Enforcement is subject to configured policy, legal requirements, and human oversight.

5.4.5 Authorization and Release
• **Function:** Only tones that satisfy validation criteria and required authorizations are released to the System Layer for execution.
• **Provenance:** Validation decisions include provenance metadata and are logged for audit and dispute resolution.

Safeguards: AURORA is designed to **support** consistent interpretation and governance; it requires governance processes for rule updates, audits, and legal review.

5.5 System Layer (Execution Layer)
Overview: The System Layer executes actions derived from validated and authorized tones. Execution is constrained by the scope, constraints, and authorizations encoded in the tone and by organizational and legal policies.
Execution Controls
• **Scope enforcement:** The System Layer enforces the scope and constraints specified in the validated tone and rejects or quarantines requests that exceed authorization.
• **Human authorization:** High-risk or legally sensitive actions require explicit human authorization according to configured policies.
• **External interactions:** Interactions with external systems follow configured interfaces, rate limits, and data-sharing policies.
Observability and Logging
• **Function:** Execution events are logged with provenance, actor identifiers, timestamps, and rationale.
• **Access controls:** Logs are retained and accessed according to retention and privacy policies to support audits and dispute resolution.
Limitations: The System Layer executes only **validated and authorized** tones; it does not act on raw human input. Implementation must include safeguards for failure modes and human override.

5.6 Responsibility and Accountability Layer (s.net)
Overview: The Responsibility Layer provides mechanisms to support clear **accountability, auditability, and escalation** when tones are ambiguous, unsafe, or otherwise require human judgment. It **supports assignment of accountability** in accordance with organizational policy and applicable law.
5.6.1 System Support for Decision Workflows
1. **Function:** Implements automated safeguards (blocking, clarification prompts, suggested rewrites) and orchestrates escalation to human decision-makers when required by policy or risk thresholds.
2. **Scope:** Automated actions are configurable and documented; high-risk or legally sensitive actions require explicit human authorization.
5.6.2 Accountability and Audit Trail
1. **Logging:** Every tone validation, decision, and execution step is logged with provenance metadata, actor identifiers (human or automated), timestamps, and rationale.
2. **Retention and access:** Logs are retained and made accessible according to retention, privacy, and audit policies.
3. **Explainability:** Records include explainability artifacts (confidence scores, rule versions, and relevant policy references) to support review and dispute resolution.
5.6.3 Operator Protection and Shared Responsibility
1. **Design goal:** Protect operators from undue attribution for ambiguous inputs by providing clarification workflows and documented decision traces.
2. **Allocation of responsibility:** The system **supports** transparent allocation of responsibility; actual legal or moral responsibility is determined by applicable law, contractual terms, and organizational governance.
3. **Dispute handling:** Includes procedures for review, correction, and remediation when outcomes are contested.
5.6.4 Compliance, Oversight, and Continuous Monitoring
1. **Compliance:** The layer enforces configurable compliance checks and integrates with legal and policy review processes.
2. **Monitoring:** Includes continuous monitoring for drift, bias, and safety incidents, and supports periodic audits and reporting.
3. **Human oversight:** Establishes clear escalation paths and human roles for oversight, review, and remediation.
Safeguards: The Responsibility Layer emphasizes **supporting accountability and human oversight** rather than asserting legal liability on behalf of the system. Implementations must include legal review and governance processes to define how responsibility is assigned in practice.

5.7 Misinterpretation Firewall

Overview: The Misinterpretation Firewall is a configurable detection and mitigation mechanism that reduces the risk of unintended or unsafe actions resulting from ambiguous or low-confidence interpretations.

Detection

1. **Function:** Uses confidence thresholds, provenance checks, and contextual heuristics to detect ambiguous or potentially spurious meanings.
2. **Metrics:** Detection performance is monitored using measurable metrics (false positive/negative rates) and subject to periodic review.

Mitigation

1. **Behaviour:** On detection, the firewall may block execution, issue clarification prompts, suggest constrained rewrites, or escalate to human review according to policy and risk tier.
2. **Human-in-the-loop:** For medium and high-risk cases, human adjudication is required before execution proceeds.

Transparency and Monitoring

1. **Logging:** All firewall decisions are logged with rationale and supporting metadata.
2. **Continuous improvement:** Detection rules and thresholds are periodically audited and updated to reduce bias and improve accuracy.

Limitations: The firewall **reduces** but does not guarantee elimination of incorrect inferences; it is part of a layered risk-management approach.

5.8 Multi-Layer Flow (Pipeline Summary)

Pipeline Summary

- **Operator expresses intent.** Unstructured input is captured with provenance metadata.
- **PDT interprets intent and generates a tone.** Outputs include confidence scores, provenance, and required authorizations.
- **AURORA validates the tone.** Validation uses versioned rules, policy checks, and risk thresholds; ambiguous or high-risk tones follow clarification or escalation workflows.
- **System executes validated and authorized tones.** Execution is constrained to the tone's scope and applicable policies; high-risk actions require human authorization.
- **Responsibility layer records provenance and supports accountability.** Logs, explainability artifacts, and escalation records support audit and dispute resolution.

Design principle: The pipeline emphasizes **transparency, configurable human oversight, and policy-bound automation** rather than absolute automation or legal assertions of system liability.

SECTION 6 – FLOW DIAGRAMS

Primary Intent-to-Action Flow

Operator expresses intent → PDT captures raw intent (provenance) → PDT interprets → generates tone (confidence, metadata) → Tone sent to AURORA → AURORA validates or routes to clarification/escalation → System executes validated tone (if authorized) → Responsibility layer logs decision and execution.

Misinterpretation Firewall Flow

PDT detects ambiguity → PDT flags tone → AURORA blocks or routes to clarification → Operator clarifies or human adjudicator resolves → PDT regenerates tone → AURORA validates → System executes → Responsibility layer logs event.

Identity Mediation Flow

Operator expresses intent → PDT determines identity scope → PDT generates scoped identity assertions using data minimization/privacy techniques → AURORA validates proof → System receives scoped identity → Responsibility layer logs identity boundaries.

Context Drift Prevention Flow

PDT generates tone with metadata → AURORA compares metadata across contexts → If drift detected, block + clarify or escalate → If no drift, approve → System executes → Responsibility layer logs drift events.

Responsibility Escalation Flow

System receives unclear request → System forwards to AURORA → AURORA identifies ambiguity → AURORA blocks execution and initiates clarification/escalation → PDT mediates and produces a clean tone → Execution proceeds if authorized → Responsibility layer logs accountability.

Multi-Device Continuity Flow
Operator switches device → PDT synchronizes semantic state → AURORA ensures continuity → Meaning preserved → System continues execution.

SECTION 7 – CLAIMS

7.1 Independent Claim
Claim 1. A method for translating human intent into machine-executable actions, comprising:
• receiving natural, unstructured human intent at an operator layer;
• interpreting the intent via a semantic translator (PDT) to generate a structured meaning unit ("tone") that includes provenance and confidence metadata;
• validating the tone using a governance substrate (AURORA) that applies versioned interpretive rules and policy checks and that routes ambiguous or high-risk tones to clarification or human review according to configurable risk thresholds;
• executing the validated and authorized tone within a digital system constrained to the scope and constraints encoded in the tone; and
• recording provenance, decision rationale, and audit logs through a responsibility and accountability layer (s.net) that supports escalation to human actors and allocation of accountability according to applicable law and organizational policy.

7.2 Dependent Claims
Claim 2 – Semantic Interpretation. The method of claim 1, wherein the PDT extracts intent, context, scope, constraints, confidence scores, and provenance.
Claim 3 – Ambiguity Detection. The method of claim 1, wherein ambiguous or low-confidence tones are routed to a configurable clarification workflow that may prompt the operator, suggest constrained rewrites, or escalate to human review.
Claim 4 – Governance Substrate. The method of claim 1, wherein the governance substrate enforces versioned interpretive rules, policy checks, and risk-based human-in-the-loop escalation to mitigate semantic drift.
Claim 5 – Identity Mediation. The method of claim 1, wherein the PDT generates scoped identity assertions that disclose only attributes required for authorization using privacy-preserving techniques and configurable data minimization policies.
Claim 6 – Responsibility Support. The method of claim 1, wherein the responsibility layer records provenance and supports allocation of accountability in accordance with applicable law and organizational policy.
Claim 7 – Multi-Device Continuity. The method of claim 1, wherein tones retain semantic consistency across devices through synchronized semantic state and metadata.
Claim 8 – Context Drift Prevention. The method of claim 1, wherein the governance substrate blocks or routes for clarification tones whose meaning shifts across contexts.
Claim 9 – Clarification Loop. The method of claim 1, wherein ambiguous or low-confidence tones are routed to a configurable clarification workflow that may prompt the operator for confirmation or additional detail, suggest constrained rewrites, or escalate to human review according to risk and policy thresholds.
Claim 10 – Scoped Execution. The method of claim 1, wherein execution of actions is limited to the scope, constraints, and authorizations defined in the validated tone and subject to applicable organizational policies and legal requirements.

7.3 System Claims
Claim 11 – Multi-Layer Semantic Mediation System. A multi-layer semantic mediation system configured to translate human intent into machine-executable actions, the system comprising:
• an **operator layer** for receiving unstructured operator input;
• a **semantic translator** (PDT) configured to generate structured meaning units ("tones") with provenance and confidence metadata;
• a **governance substrate** (AURORA) configured to validate tones against interpretive rules, policy, and risk thresholds and to coordinate clarification and escalation workflows;
• a **system execution layer** that executes only validated and authorized tones within the scope and constraints encoded in the tones; and

• a **responsibility and accountability layer** (s.net) that records provenance, decision rationale, and audit logs and that orchestrates escalation to human actors when required.

Claim 12 – Translator Layer. The system of claim 11, wherein the semantic translator generates structured tones that include intent, contextual metadata, scope, constraints, confidence scores, and provenance information, and is configurable to surface explainability and bias-monitoring artifacts.

Claim 13 – Governance Layer. The system of claim 11, wherein the governance substrate validates tones using versioned interpretive rules, configurable policy checks, and risk-based human-in-the-loop escalation; the governance substrate is further configured to mitigate semantic drift through monitoring and human review.

Claim 14 – Accountability Layer. The system of claim 11, wherein the responsibility layer logs validation and execution events with provenance metadata, actor identifiers, timestamps, and rationale; the responsibility layer supports audit, dispute resolution, and configurable retention and access controls in accordance with applicable policies and law.

7.4 Computer-Readable Medium Claim

Claim 15. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause the processors to perform the method of claim 1 as implemented with the configurable clarification workflows, scoped execution controls, audit logging, and policy-based escalation described herein, subject to applicable legal and organizational constraints.

Implementation and Governance Notes (Concise)

• **Avoid absolutes:** Replace "never/always/cannot" with qualified, policy-bound language.

• **Human oversight matrix:** Define which actions are automated and which require human authorization, mapped to risk tiers.

• **Privacy controls:** Document data minimization, purpose limitation, consent flows, retention, and privacy-preserving techniques.

• **Bias and safety testing:** Include bias monitoring, fairness testing, and periodic audits.

• **Legal review:** Require legal and ethics review before deployment in regulated contexts.

• **Auditability:** Ensure logs include provenance, rule versions, actor identifiers, timestamps, and rationale.

AETHENA INSTITUTE

COMPLIANCE APPENDIX — Privacy, Consent, Retention, Audit, and Human-Oversight Matrix

1 Privacy and Data Minimization

• **Purpose limitation:** Data collected by the PDT and related layers is processed only for the purpose of intent interpretation, authorization, execution, audit, and safety monitoring.
• **Data minimization:** Only attributes strictly necessary for authorization or audit are requested or stored; identity mediation uses scoped assertions and selective disclosure where possible.
• **Privacy-preserving techniques:** Implementations should support selective disclosure, pseudonymization, and cryptographic proofs (e.g., selective disclosure tokens) when feasible.
• **Operator transparency:** Operators are informed which attributes are requested and why; consent and revocation options are provided consistent with applicable law.
• **Legal constraints:** All processing is subject to applicable data protection laws and contractual obligations.

2 Consent and Notice

• **Notice at point of capture:** Provide clear, concise notice when collecting identity attributes or sensitive contextual data, including purpose, retention, and contact for queries.
• **Granular consent:** Obtain consent for identity attributes and for any secondary uses beyond immediate intent translation (e.g., analytics, model improvement). Consent records are logged.
• **Revocation:** Operators can revoke consent where legally permitted; revocation triggers revalidation of any dependent authorizations and may require re-authentication.
• **Automated decision disclosure:** When automated workflows materially affect rights or outcomes, provide accessible explanations and an option for human review.

3 Retention and Access Controls

• **Retention policy:** Define retention periods for: raw inputs, interpreted tones, validation logs, execution logs, and audit trails. Example default categories:
• **Raw inputs:** [e.g., 7–30 days] unless required for investigation.
• **Validated tones and provenance:** [e.g., 1–3 years] for audit and dispute resolution.
• **Execution logs and rationale:** [e.g., 3–7 years] or as required by law.
• **Access controls:** Role-based access to logs and identity data; least privilege enforced; access requests and disclosures are logged.
• **Data deletion:** Implement secure deletion procedures and document exceptions (e.g., legal hold).
• **Data export and portability:** Provide mechanisms for authorized export consistent with law and policy.

4 Auditability and Explainability

• **Provenance metadata:** Every tone, validation decision, and execution event includes timestamp, actor identifier (human or automated), rule/version used, confidence score, and rationale summary.
• **Explainability artifacts:** Store rule versions, model configuration, and confidence metrics sufficient to reconstruct decisions for audits and dispute resolution.
• **Audit logs:** Immutable or tamper-evident logs; retention and access governed by policy.
• **Periodic audits:** Schedule regular audits for drift, bias, safety incidents, and compliance; produce audit reports and remediation plans.
• **Dispute resolution:** Maintain a documented process for operators to request review, correction, or remediation of outcomes.

5 Human-Oversight Matrix (Risk-Based)

Risk Tier	Automated Actions Allowed	Human Authorization Required	Escalation Path	Retention Example
Low	Routine, reversible actions within narrow scope; auto-confirmations.	No	Operator prompt → PDT clarification → AURORA auto-approve.	30–90 days
Medium	Actions with moderate impact; may be auto-suggested but gated.	Conditional (if confidence < threshold or policy flag)	AURORA → Operator confirmation → Human reviewer if unresolved.	1–3 years
High	High-impact, irreversible, or legally sensitive actions.	**Yes, explicit human sign-off required**	AURORA → Human adjudicator → Legal/compliance review if needed.	3–7 years or per law

Notes:
- **Confidence thresholds** and policy flags are configurable per deployment.
- **Escalation contacts** (roles, not names) must be defined in governance documentation.
- **Auditability** is required at every tier; automated actions must record rationale and rule versions.

6 Operational Controls and Governance

- **Rule governance:** Maintain a rule registry with versioning, authorship, test results, and deployment history. Changes require documented review and rollback procedures.
- **Bias and fairness testing:** Integrate pre-deployment and periodic fairness tests; log test results and corrective actions.
- **Monitoring and alerting:** Implement real-time monitoring for drift, anomaly detection, and safety incidents; define SLAs for incident response.
- **Legal and ethics review:** Require legal and ethics sign-off for rule sets that affect rights, safety, or regulated domains.
- **Training and roles:** Define roles for operators, reviewers, auditors, and incident responders; provide training on system limits and escalation procedures.

7 Minimal Contractual and Policy Clauses to Include

- **Limitation of automation:** State that automation is policy-bound and that legal responsibility is determined by law and contractual terms.
- **Data processing addendum:** Specify processors, subprocessors, purposes, retention, and security measures.
- **Audit rights:** Define auditor access, frequency, and scope.
- **Liability and indemnity:** Clarify allocation of liability for misuse, breaches, or failures, consistent with organizational risk posture.

8 Implementation Checklist (Quick)

- Configure **confidence thresholds** and map to the Human-Oversight Matrix.
- Publish **retention schedules** and implement secure deletion.
- Enable **immutable provenance logging** and role-based access.
- Run **bias and safety tests** before deployment and periodically thereafter.
- Document **escalation contacts** and dispute resolution procedures.
- Obtain **legal and ethics sign-off** prior to production use.